Notes: Bandiera, Barankay & Rasul (Econometrica, 2009)

Social Connections and Incentives in the Workplace: Evidence From Personnel Data

Contents

1 Big picture

- **Question.** Do social connections between managers and workers improve productivity by helping managers motivate workers, or do they reduce firm performance by creating favoritism?
- **Empirical setting.** The paper studies a soft-fruit farm in the United Kingdom using detailed personnel data on seasonal Eastern European workers and managers. Workers pick fruit and are paid piece rates; managers organize logistics and can target effort to specific workers.
- **Main intervention.** During the 2003 season, the authors engineered a natural field experiment that changed managerial compensation from fixed wages to fixed wages plus a performance bonus based on the average productivity of workers on the field that day. Workers' piece-rate incentives did not change.
- **Key social-connection measures.** A worker and manager are classified as socially connected if they share nationality, live near each other on the farm, or arrived at the farm in a similar cohort. These proxies are validated using later friendship-survey evidence.
- **Core identification idea.** The same worker is observed on field-days when he is socially connected to managers and on field-days when he is not; the same manager works with both connected and unconnected workers; and the strength of managerial incentives changes exogenously.
- **Main worker-level result.** Under fixed manager wages, connected workers are more productive. Using the share of managers connected to the worker, being connected to all managers rather than none raises productivity by about 9% relative to the baseline productivity of unconnected workers.
- **Main incentive-interaction result.** Once managers are paid performance bonuses, the productivity advantage of connected workers disappears. Managers stop favoring connected workers and instead direct effort toward higher ability workers.
- **Allocation result.** Social connections increase the level of managerial effort toward connected workers, but when incentives are weak they distort the allocation of effort away from high ability workers and toward connected low ability workers.
- **Firm-level result.** Average field-day productivity falls when social connectedness is higher under fixed wages. The estimated reduction is around 5–8% for a one-standard-deviation increase in average connections, depending on controls.
- **Economic takeaway.** Social ties within firms are not inherently good or bad. Their effect depends on whether they improve communication and motivation or distort managerial allocation. In this farm setting, where tasks are simple and effort can be targeted, favoritism dominates under low-powered incentives.

2 Incremental contribution of the paper

- **First quantitative evidence on cross-tier workplace social connections.** Earlier work studied social relations among workers or among managers. This paper studies vertical ties between managers and subordinates and shows that such ties directly shape productivity and allocation of effort.
- **Combines personnel data with an internally implemented field experiment.** The authors do not rely only on observational variation in friendships. They combine worker-day productivity records with an exogenous change in managerial incentives, which allows them to test whether managers change their treatment of connected workers when their objectives are aligned with the firm.
- **Separates the level effect from the allocation effect of social connections.** Social connections can increase effort toward connected workers, but they may allocate managerial help to the wrong workers. The paper formalizes and empirically estimates both forces.
- **Introduces a revealed-preference test of favoritism.** If managers reallocate effort away from connected low ability workers and toward high ability workers when bonuses are introduced, their previous allocation under fixed wages could not have been productivity maximizing.

- **Measures social ties without survey-reported friendship in the main sample.** The paper uses nationality, arrival cohort, and living proximity as ex ante connection proxies and then validates them with later friendship survey evidence.
- **Shows why incentive design and social organization are complements.** Monetary incentives do not merely increase effort; they change whom managers choose to help. This links organizational social structure to optimal compensation design.
- **Provides an efficiency benchmark for favoritism.** The paper does not define favoritism simply as helping friends. It asks whether helping connected workers lowers average productivity relative to the allocation chosen when managers face stronger incentives.
- **General lesson.** Social capital inside firms may improve productivity in tasks requiring communication and problem solving, but it can reduce productivity in simple measurable tasks when managers can redirect help toward socially connected workers.

3 Data and measurement

3.1 Workplace setting

- The firm is a leading U.K. producer of soft fruit.
- Workers pick fruit; managers organize logistics such as row assignment, crate replacement, and reallocations across rows.
- Workers are paid piece rates, so worker earnings move one-for-one with productivity.
- Managers can raise a worker's productivity by targeting logistical help, but effort is scarce and can be redirected across workers.
- The analysis focuses on the main fruit, the main farm site, and the peak picking season from May 1 to August 31, 2003.

Interpretation: The production process is ideal for measuring favoritism because managers have observable actions that affect individual productivity, and productivity is measured precisely in kilograms picked per hour.

3.2 Managerial incentive change

- Before June 27, 2003, managers are paid fixed wages.
- After June 27, managers receive fixed wages plus a performance bonus based on average worker productivity on the field-day.
- The performance bonus is paid only if fruit quality does not deteriorate, limiting multitasking concerns.
- The bonus increases expected managerial earnings meaningfully: conditional on receiving the bonus, hourly earnings rise by about 25%; averaged across all post-bonus field-days, earnings rise by about 7%.

Interpretation: The managerial bonus raises the private return to maximizing the field's average output. This creates the core treatment variation used to test whether previous allocation decisions were distorted by social ties.

3.3 Worker productivity

- The dependent variable is log worker productivity on a worker-field-day.
- Productivity is kilograms of fruit picked per hour.
- The data are recorded electronically, reducing measurement error.
- Worker ability is proxied by average productivity under the bonus regime, when managers have stronger incentives to allocate effort productively.

Interpretation: Because workers are paid piece rates, productivity is also a direct measure of worker earnings. Managerial favoritism therefore redistributes both productivity and income across workers.

3.4 Social connection measures

A worker-manager pair is socially connected if any of the following hold:

- **Same nationality:** the worker and manager come from the same country.
- **Same arrival cohort:** their identification numbers fall within the same 10-digit window, proxying for similar arrival time.
- **Same living area:** they live within five caravan numbers of each other on the farm.

For each worker-field-day, the main continuous connection measure is

$$C_{ift} = \frac{\sum_{j \in M_{ft}} c_{ij}}{M_{ft}},$$

where $c_{ij} = 1$ if worker i is connected to manager j, and M_{ft} is the number of managers on field f on day t.

Interpretation: The continuous measure is useful because a worker can be connected to zero, one, or multiple managers on a field-day. It captures the intensity of potential favoritism.

3.5 Panel structure and sample

- Main worker-level regressions use 8,884 worker-field-day observations.
- The identifying sample is restricted to workers observed at least one week under each incentive scheme and connected to at least one manager on at least one field-day.
- Table I reports 5,137 observations under fixed wages and 3,747 under performance bonuses.
- Field-day regressions use 241 field-day observations.

Interpretation: The estimation compares the same workers under different connection states and incentive regimes, rather than comparing intrinsically connected workers to intrinsically unconnected workers.

4 Theoretical framework

4.1 Production technology

There is one manager and two workers: a high-ability worker h and a low-ability worker l. Worker i's output is

$$y_i = \theta_i k_i m_i,$$

where $\theta_h = \theta > 1$, $\theta_l = 1$, m_i is managerial effort targeted to worker i, and k_i captures complementarity between manager effort and worker productivity.

Interpretation: Managerial help is productive, but it is targeted. Helping one worker means not helping another worker, so the allocation of managerial effort matters for total output.

4.2 Manager pay and social preferences

Workers are paid piece rates:

$$p_i^W = y_i.$$

Managers are paid

$$p^M = f + bY, \qquad Y = \sum_i y_i,$$

where b measures the strength of managerial performance incentives. The manager's utility is

$$u^M = p^M + \sum_{i \in \{h,l\}} \sigma_i p_i^W.$$

If the manager is socially connected to worker i, then $\sigma_i = \sigma > 0$; otherwise $\sigma_i = 0$.

Interpretation: The manager may care about a connected worker's pay because of altruism, reciprocal obligations, social rewards, or kickbacks. The model is deliberately reduced form.

4.3 Manager effort allocation problem

The manager chooses targeted effort (m_h, m_l) to maximize

$$(b + \sigma_h)\theta k(\sigma_h) m_h + (b + \sigma_l) k(\sigma_l) m_l - C(m_h + m_l).$$

Interpretation: The marginal benefit of helping worker i has two components: the monetary incentive b and the social weight σ_i. Increasing b pushes the manager toward productivity-maximizing allocation; increasing σ_i pushes the manager toward connected workers.

4.4 Proposition 1: level and allocation effects

- Social connections weakly increase managerial effort because they raise the manager's private return to helping connected workers.
- Social connections may change the allocation of effort toward the connected worker.
- If the connected worker is high ability, this can be efficient.
- If the connected worker is low ability, this can reduce firm productivity.

Interpretation: Social ties have a positive motivational component and a potentially negative allocation component. The empirical analysis is designed to sign the allocation component.

4.5 Revealed-preference test for negative allocation

The model implies the following test:

- Under fixed wages, managers should target connected workers, regardless of worker ability.
- Under performance bonuses, managers should target high ability workers, regardless of connection status.
- If low ability connected workers lose productivity after the bonus, and high ability unconnected workers gain productivity, then the pre-bonus allocation was not productivity maximizing.

Interpretation: The bonus does not only raise effort. It changes the identity of the workers who receive managerial help. This is the key evidence of favoritism.

5 Models and empirical specifications

5.1 Baseline worker-level regression

The main regression is

$$y_{ift} = \alpha_i + \lambda_f + \gamma_0(1 - B_t)C_{ift} + \gamma_1 B_t C_{ift} + \rho B_t + \sum_k \sum_{d \in N_k} \tau_d^k (B_t D_{id}^k) + \sum_{s \in M_{ft}} \mu_s S_{sft} + \delta X_{ift} + \eta Z_{ft} + \phi t + u_{ift}.$$

- y_{ift} is log productivity of worker i on field f and day t.
- B_t equals one after the managerial bonus is introduced.
- C_{ift} is worker-manager social connectedness on the field-day.
- α_i are worker fixed effects.
- λ_f are field fixed effects.
- S_{sft} are manager indicators.
- X_{ift} captures worker picking experience.
- Z_{ft} captures the field life cycle.

Interpretation: γ_0 measures the effect of social connections when managers are paid fixed wages. γ_1 measures the effect under performance bonuses. The comparison $\gamma_0 - \gamma_1$ is the key difference-in-differences estimate.

5.2 Heterogeneous worker effects and field-date fixed effects

The paper adds two important robustness variations:

- Worker fixed effects interacted with the performance bonus, allowing each worker to respond differently to the bonus regime.
- Field-date fixed effects, absorbing any field-day shocks such as field condition, piece rate, weather, worker mix, or manager lobbying.

Interpretation: These specifications sharpen identification by asking whether, within the same field-day, the more connected workers perform differently from their own average connection-adjusted productivity.

5.3 Quantile regressions

To study allocation across ability levels, the paper estimates

$$Q_\theta(y_{ift}|\cdot) = \alpha_\theta B_t + \beta_\theta C_{ift} + \gamma_\theta B_t C_{ift} + \lambda_{\theta f} + \sum_{s \in M_{ft}} \mu_{\theta s} S_{sft} + \delta_\theta X_{ift} + \eta_\theta Z_{ft} + \phi_\theta t.$$

Interpretation: If the bonus reallocates managerial effort toward high ability workers, the bonus should reduce productivity at low quantiles for connected workers but raise productivity at high quantiles, especially for previously unconnected workers.

5.4 Field-day average productivity regression

To estimate the net effect on firm output, the paper aggregates to the field-day level:

$$\bar{y}_{ft} = \kappa_0(1 - B_t)\bar{C}_{ft} + \kappa_1 B_t \bar{C}_{ft} + \rho B_t + \sum_{s \in M_{ft}} \mu_s S_{sft} + \eta Z_{ft} + \phi t + \lambda_f + u_{ft}.$$

Interpretation: This specification asks whether the positive effect on connected workers is large enough to offset any misallocation away from more productive workers. The answer is no under fixed wages.

6 Identification and empirical design

6.1 Source of variation

- Daily worker-manager assignment varies because the COO assigns workers and managers to fields based on production needs.
- The same worker is observed when connected and when not connected.
- The same manager is observed with connected and unconnected workers.
- The managerial compensation scheme changes exogenously through the authors' field experiment.

Interpretation: The design uses both within-worker variation and an experimental change in incentives. This is much stronger than a cross-sectional comparison of connected and unconnected people.

6.2 Main identifying assumptions

- Unobserved determinants of worker-manager allocation are orthogonal to the incentive regime.
- Any connection effect unrelated to the incentive regime would have evolved similarly over time.
- Worker composition changes over the season do not drive the results.
- Field-day shocks are not systematically correlated with worker connection status in a way that changes at the bonus date.

Interpretation: The paper directly tests these concerns using field-date fixed effects, worker-by-bonus fixed effects, placebo bonus dates, 2004 placebo data, and the COO allocation algorithm.

6.3 Why the experiment is a natural field experiment

- Managers were unaware that the compensation change was part of a scientific experiment.
- The firm recorded productivity data in ordinary business operations.
- The bonus was an actual compensation policy, not a hypothetical task.
- Workers' own piece-rate incentives remained unchanged.

Interpretation: The setting combines real stakes, real production, and high-frequency individual productivity data.

6.4 Threats and paper responses

- **Sorting by the COO.** Table A.I shows social connections do not predict selection into picking or unemployment differentially by incentive regime.
- **Seasonality.** Table A.III adds farm, field, and worker time trends and placebo dates.
- **Field-day shocks.** Table III includes field-date fixed effects.
- **Heterogeneous worker responses.** Table III includes worker fixed effect by bonus interactions.
- **Connection proxy validity.** Appendix evidence separates nationality, living area, and arrival cohort; results are similar across observable and less observable dimensions.

Interpretation: The empirical case is strongest because several threats would predict the same connection effect under both incentive regimes, but the estimated connection effect disappears under performance pay.

7 Section-by-section study guide

- **Introduction.** Focus on why social connections are theoretically ambiguous and why managerial incentives help reveal whether favoritism is costly.
- **Empirical setting.** Understand how workers are paid, how managers help workers, and why manager effort can be targeted.
- **Theoretical framework.** Track the distinction between level effects and allocation effects.
- **Data and descriptives.** Use Tables I and II to see how connection measures vary and why raw productivity patterns motivate the later regressions.
- **Worker-level regressions.** Table III is the core causal evidence that social connections matter under fixed wages but not under performance pay.
- **Heterogeneity and quantiles.** Table IV and Figure 1 show how effort moves across the productivity distribution after the bonus.
- **Firm-level effect.** Table V answers the welfare question for the firm: under fixed wages, social connections reduce average productivity.
- **Appendix.** The appendix provides evidence that the main estimates are not driven by COO allocation, connection definitions, or time effects.

8 Tables and figures

8.1 Table I: Descriptives on the Social Connectivity Between Workers and Managers by Managerial Incentive Scheme

Read:

- Average connectedness is very similar under the two regimes: 0.425 under fixed wages and 0.412 under performance bonuses.
- Same nationality connections are the largest component, with averages of 0.304 and 0.285.
- Living area connections are around 0.139 and 0.138.
- Arrival cohort connections are small on average, 0.038 and 0.056, but valuable because they are harder for the COO to observe and manipulate.
- The table reports both between-worker and within-worker standard deviations, showing substantial within-worker variation in social connectedness.

Detailed interpretation: The near equality of connectedness before and after the bonus suggests that the change in managerial incentives is not mechanically associated with a change in assignment to connected managers. The within-worker variation is crucial: the paper identifies effects by comparing the same worker across connected and unconnected field-days.

Economic message: Table I establishes that connection exposure is not a fixed worker trait. It varies enough within worker to identify social favoritism, and it is balanced across incentive regimes.

TABLE I

DESCRIPTIVES ON THE SOCIAL CONNECTIVITY BETWEEN WORKERS AND MANAGERS BY MANAGERIAL INCENTIVE SCHEME[a]

	Managerial Incentive Scheme	
Share of Managers	Fixed Wages	Performance Bonus
Connected to i (C_{ift})	.425	.412
	(.297)	(.300)
	[.196]	[.154]
Who are the same nationality as i	.304	.285
	(.344)	(.319)
	[.145]	[.111]
Who are in the same living area as i	.139	.138
	(.116)	(.172)
	[.167]	[.138]
Who are from the same arrival cohort as i	.038	.056
	(.076)	(.101)
	[.079]	[.074]

[a] All variables are defined at the worker-field-day level. Means, standard deviation between workers is in parentheses, and standard deviation within worker is in brackets. A manager and worker are defined to be resident in the same living area if they live within five caravans from each other on the farm. A manager and worker are defined to be in the same arrival cohort if they have identification numbers within the same 10 digit window. A manager and given worker i are defined to be connected if they are either of the same nationality, live in the same area, or are in the same arrival cohort. The sample is restricted to the 129 workers who work for at least one week under both incentive schemes and are connected to at least one manager on at least one dimension. On average, each worker is

8.2 Table II: Worker Productivity by Social Connectivity to Managers and Managerial Incentive Scheme

Read:

- Among all workers, connected workers produce 8.98 kg/hr under fixed wages versus 7.21 kg/hr when unconnected, a difference of 1.77 kg/hr.
- Under performance bonuses, the connected-unconnected difference for all workers falls to 0.179 and is not significant.
- Among low ability workers, connected workers produce more under fixed wages, but their productivity is slightly lower under performance pay.
- Among high ability workers, productivity rises sharply under performance bonuses, especially for workers who are not socially connected to managers.
- The raw difference-in-difference patterns already suggest that the bonus redirects managerial help away from low ability connected workers and toward high ability workers.

Detailed interpretation: Table II is descriptive, not the final causal design, but it summarizes the paper's mechanism. Fixed-wage managers appear to help connected workers regardless of ability. Bonus-paid managers appear to help high ability workers even if they are not socially connected. The table foreshadows the allocation effect: the same social connection that raises an individual worker's productivity can lower total productivity if it targets the wrong workers.

Economic message: Social connections raise connected workers' earnings when managers are not paid for productivity, but this gain reflects favoritism rather than firm-value-maximizing effort allocation.

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	(.116)	(.172)
	[.167]	[.138]
Who are from the same arrival cohort as i	.038	.056
	(.076)	(.101)
	[.079]	[.074]

[a]All variables are defined at the worker-field-day level. Means, standard deviation between workers is in parentheses, and standard deviation within worker is in brackets. A manager and worker are defined to be resident in the same living area if they live within five caravans from each other on the farm. A manager and worker are defined to be in the same arrival cohort if they have identification numbers within the same 10 digit window. A manager and given worker i are defined to be connected if they are either of the same nationality, live in the same area, or are in the same arrival cohort. The sample is restricted to the 129 workers who work for at least one week under both incentive schemes and are connected to at least one manager on at least one dimension. On average, each worker is

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TABLE II

WORKER PRODUCTIVITY (KG/HR), BY SOCIAL CONNECTIVITY TO MANAGERS AND MANAGERIAL INCENTIVE SCHEME[a]

	Managerial Incentive Scheme:		
	Fixed Wages	Performance Bonus	Difference
Panel A: All Workers			
Unconnected on field-day ($DC_{ift} = 0$)	7.21	9.52	2.30***
	(.211)	(.600)	(.535)
Connected on field-day ($DC_{ift} = 1$)	8.98	9.70	.712
	(.345)	(.527)	(.272)
Difference	1.77***	.179	1.596***
	(.352)	(.750)	(.609)
Panel B: Low Ability Workers			
Unconnected on field-day ($DC_{ift} = 0$)	6.10	6.60	.506
	(.248)	(.342)	(.423)
Connected on field-day ($DC_{ift} = 1$)	7.37	6.77	−.603**
	(.173)	(.212)	(.211)
Difference	1.27***	.161	1.11**
	(.287)	(.368)	(.435)
Panel C: High Ability Workers			
Unconnected on field-day ($DC_{ift} = 0$)	7.76	10.79	3.03***
	(.259)	(.672)	(.609)
Connected on field-day ($DC_{ift} = 1$)	10.32	11.61	1.28***
	(.519)	(.668)	(.338)
Difference	2.56***	.815	1.75**
	(.518)	(.899)	(.719)

[a]Means, standard errors are given in parentheses. *** denotes significance at 1%, ** at 5%, and * at 10%. All variables are defined at the worker-field-day level. The standard errors on the differences and difference-in-difference are estimated from running the corresponding least squares regression, allowing the standard errors to be clustered by worker. Productivity is measured as the number of kilograms of fruit picked per hour by the worker on the field-day. A manager and given worker i are defined to be connected if they are either of the same nationality, live in the same area, or are in the same arrival cohort. A worker is defined to be unconnected on the field-day if she is not socially connected to any of her managers that field-day. A worker is defined to be connected on the field-day if she is socially connected to at least one of her managers. Low (high) ability workers are those whose average productivity under the bonus is below (above) the median average productivity.

8.3 Table III: Social Connections and Managerial Incentives

Read:

- Column 1 uses a binary indicator for whether any manager is connected to the worker. Under fixed wages, connection raises productivity by about 4.9%.
- Column 2 uses the continuous share of connected managers. Under fixed wages, the coefficient is 0.158 and highly significant.

- Under performance bonuses, the connection coefficient is negative but insignificant in the continuous specification.
- The difference-in-difference estimate in column 2 is 0.241 and significant at the 1% level, indicating a sharp decline in the connection effect when bonuses are introduced.
- Column 3 allows heterogeneous worker responses to the bonus, and column 4 adds field-date fixed effects; both preserve the central pattern.
- Adjusted R^2 rises to 0.5817 with field-date fixed effects, showing that field-day factors matter, but they do not explain away the social-connection effect.

Detailed interpretation: This is the main worker-level causal table. The effect of connections under fixed wages is identified from within-worker variation in connection intensity. The disappearance of the effect under performance pay is the key evidence that managers change behavior when their incentives align with firm output. The field-date fixed-effect specification is especially demanding because it compares workers on the same field and day, exposed to the same weather, field quality, piece rate, and managers.

Economic message: Low-powered manager incentives create space for favoritism. High-powered incentives suppress it by making average field productivity valuable to managers.

TABLE III

SOCIAL CONNECTIONS AND MANAGERIAL INCENTIVES[a]

	1. Any Managers Connected To	2. Share of Managers Connected To	3. Heterogeneous Effects of the Bonus on Workers	4. Field-Date Fixed Effects
Any managers connected to i, fixed wages for managers (DC_{ift})	.049**			
	(.019)			
Any managers connected to i, performance bonus for managers (DC_{ift})	.016			
	(.032)			
Share of managers connected to i, fixed wages for managers (C_{ift})		.158***	.143***	.106**
		(.040)	(.040)	(.045)
Share of managers connected to i, performance bonus for managers (C_{ift})		−.083	−.079	−.060
		(.088)	(.088)	(.061)
Difference-in-difference estimate	.033	.241***	.222**	.167**
	(.034)	(.095)	(.096)	(.075)
Interactions of nationality × performance bonus dummy	Yes [.147]	Yes [.042]	No	No
Interactions of living site × performance bonus dummy	Yes [.000]	Yes [.000]	No	No
Interactions of arrival cohort × performance bonus dummy	Yes [.000]	Yes [.000]	No	No
Interactions of worker fixed effect × performance bonus dummy	No	No	Yes [.000]	Yes [.000]
Field-date fixed effects	No	No	No	Yes
Adjusted R-squared	.4355	.4361	.4479	.5817
Number of observations (worker-field-day)	8884	8884	8884	8884

[a]Dependent variable = log of worker's productivity (kilograms picked per hour on the field-day). *** denotes significance at 1%, ** at 5%, and * at 10%. In columns 1 to 3 the standard errors allow for clustering at the worker level. In column 4 standard errors are clustered at the field-date level. All specifications control for worker, field, and manager fixed effects. The other controls included in specifications 1 to 3 include the managerial performance bonus dummy, the worker's picking experience, the field life cycle, a time trend, and interactions between the performance bonus dummy and the worker's nationality, arrival cohort, and living site. The field life cycle is defined as the nth day the field is picked divided by the total number of days the field is picked over the season. In column 4 these interactions are replaced by interactions of the worker fixed effect and the performance bonus dummy, and a series of field-date fixed effects and hence the field life cycle and time trend are dropped from this specification. All continuous variables are in logarithms. A manager and given worker i are defined to be connected if they are either of the same nationality, live in the same area, or are in the same arrival cohort. All sample workers are connected to at least one manager on at least one field-day and work at least one week under each incentive scheme. In column 1 a worker is defined to be unconnected on the field-day if she is not socially connected to any of her managers that field-day, and the worker is defined to be connected on the field-day if she is socially connected to at least one of her managers. The difference-in-difference estimate is the difference in the effect of social connections on worker productivity by managerial incentive scheme. At the foot of each column we report the p-value on the F-test on the joint significance of the interaction terms with the performance bonus dummy.

TABLE IV

QUANTILE REGRESSION ESTIMATES[a]

	Quantile of Worker Productivity				
	10th	25th	50th	75th	90th
Performance bonus for managers (α_θ)	−.093	.001	.100**	.214***	.312***
	(.078)	(.037)	(.034)	(.040)	(.041)
Share of managers connected to i (β_θ)	.013	.074**	.144***	.272***	.392***
	(.067)	(.032)	(.030)	(.036)	(.037)
Performance bonus for managers × share of managers connected to i (γ_θ)	−.228**	−.175***	−.179***	−.168***	−.233***
	(.104)	(.050)	(.047)	(.057)	(.060)
Implied effect of performance bonus for managers on connected field-days ($\alpha_\theta + \gamma_\theta C_{ift}$) (evaluated at mean of C_{ift})	−.166**	−.055*	.042	.160***	.237***
	(.069)	(.033)	(.031)	(.035)	(.035)
Number of observations (worker-field-day)	8884	8884	8884	8884	8884

[a]Dependent variable = log of worker's productivity (kilograms picked per hour on the field-day). *** denotes significance at 1%, ** at 5%, and * at 10%. Standard errors are reported in parentheses. All specifications control for

8.4 Table IV: Quantile Regression Estimates

Read:

- The performance bonus effect is small or negative at the bottom of the productivity distribution and positive at the top.
- The fixed-wage connection effect rises sharply across the distribution, from 0.013 at the 10th percentile to 0.392 at the 90th percentile.
- The interaction of bonus and connection is negative at all quantiles, meaning the connection advantage is reduced once bonuses are introduced.
- The implied bonus effect on connected field-days is negative at the 10th and 25th percentiles, near zero at the median, and positive at higher quantiles.
- This pattern is exactly what the allocation story predicts: low ability connected workers lose support under bonuses, while high ability workers gain.

Detailed interpretation: Quantile regressions allow the effect of social ties to differ across the conditional productivity distribution. This is important because the theory is about allocation across worker ability, not just average productivity. The table shows that the bonus changes who receives managerial help. The bottom quantiles suffer on connected field-days after the bonus, while the top quantiles benefit.

Economic message: The bonus reveals that fixed-wage managers were over-supporting socially connected low productivity workers relative to the productivity-maximizing allocation.

8.5 Figure 1: Worker's Fixed Effects by Connection Status and Managerial Incentive Scheme

Read:

- Panel (a) compares connected and unconnected days under fixed wages; the connected distribution is shifted right.
- Panel (b) compares connected and unconnected days under performance bonuses; the two distributions are much closer.
- Panel (c) compares fixed-wage and bonus regimes for connected days; the left tail thickens after the bonus, consistent with low ability connected workers losing managerial support.
- Panel (d) compares fixed-wage and bonus regimes for unconnected days; the distribution shifts right for high productivity workers.
- The figure visualizes the same allocation pattern as Table IV.

Detailed interpretation: The figure is useful because it does not impose a single average effect. It shows distributional reallocation. Under fixed wages, connection status matters visibly. Under bonuses, ability matters more, and the connection premium fades. The connected-worker panel shows that performance pay can reduce output for some connected workers because managers stop targeting them.

Economic message: Managerial incentives do not merely raise effort; they reallocate effort across the workforce.

8.6 Table V: The Effect of Social Connections on Average Productivity

Read:

- The dependent variable is log average field-day productivity.
- Under fixed wages, average connections have negative and significant coefficients across specifications.
- The unconditional coefficient is -0.779; with field fixed effects it remains -0.535.
- Under performance bonuses, the average connection coefficients are small and statistically insignificant.
- The table uses field-day observations, so it directly addresses firm-level output rather than individual worker redistribution.

Detailed interpretation: Table V answers the paper's central welfare question. Tables II-IV show that connected workers benefit under fixed wages, but Table V shows that this benefit comes at the cost of lower aggregate productivity. The result implies that the positive level effect of social connections is not large enough to offset the negative allocation effect when managers face low-powered incentives.

Economic message: Social connections are costly for the firm in this setting because favoritism misallocates managerial help away from the workers where it would raise output most.

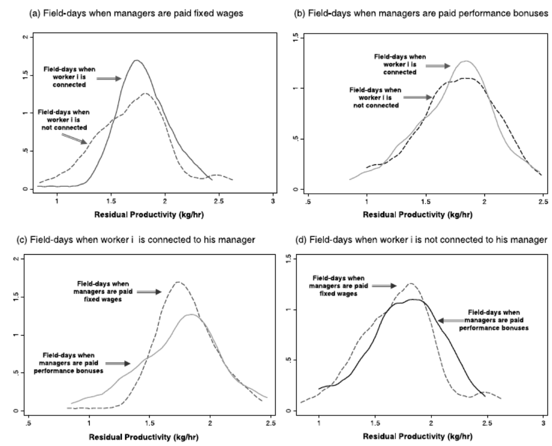

TABLE V

THE EFFECT OF SOCIAL CONNECTIONS ON AVERAGE PRODUCTIVITY

	1. Unconditional	2. Managers FE	3. Time Controls	4. Field FE
Average connections ($\bar{C}_{ift}$), fixed wages for managers	−.779***	−.544**	−.675***	−.535**
	(.201)	(.255)	(.238)	(.250)
Average connections ($\bar{C}_{ift}$), performance bonus for managers	.206	.149	.122	.031
	(.509)	(.529)	(.484)	(.480)
Manager fixed effects	No	Yes	Yes	Yes
Time controls	No	No	Yes	Yes
Field fixed effects	No	No	No	Yes
Adjusted R-squared	.944	.943	.948	.909
Number of observations (field-day)	241	241	241	241

[a] Dependent variable = log of average field-day productivity (kilograms picked per hour on the field-day). *** denotes significance at 1%, ** at 5%, and * at 10%. AR(1) regression estimates are reported. Panel corrected standard errors are calculated using a Prais–Winsten (AR(1)) regression, allowing the error terms to be field specific heteroskedastic, and contemporaneously correlated across fields. The autocorrelation process is assumed to be specific to each field. Each field-day observation is weighted by the log of the number of workers present. All specifications control for the bonus dummy that is equal to 1 when the managerial performance bonus scheme is in place and equal to 0 otherwise. The dependent variable is the log of the average productivity at the field-day level. The average connection variable is computed as the average of the social connection measure for all workers on the field-day. A manager and given worker i are defined to be connected if they are either of the same nationality, live in the same area, or are in the same arrival cohort. Time controls included in columns 3 and 4 are a linear time trend and the field life cycle, defined as the nth day the field is picked divided by the total number of days the field is picked over the season.

8.7 Table A.I: Social Connections, Managerial Incentives, and the COO's Allocation Algorithm

Read:

- The table tests whether social connections predict the probability of being selected to pick or being unemployed.
- Performance bonuses affect selection outcomes, but the key connection-by-bonus patterns do not support systematic manipulation by the COO.
- The table separates selection into picking and unemployment, addressing a central assignment concern.

Detailed interpretation: A potential threat is that connected workers are assigned to better field-days or are protected from unemployment. Table A.I is designed to test this selection channel. It shows that the main findings are not driven by a connection-based assignment algorithm that changes at the bonus date.

Economic message: The productivity results are less likely to reflect who gets selected to work and more likely to reflect how managers allocate effort once workers are on the field.

8.8 Table A.II: Social Connections and Managerial Incentives by Type of Social Connection

Read:

- The table separates nationality, living area, and arrival cohort connections.
- Under fixed wages, each connection type is positively related to productivity.
- Under performance bonuses, connection effects are not positive in the same way.
- The arrival cohort measure is particularly useful because it is less easily observable to managers or the COO than nationality.

Detailed interpretation: Similar results across connection dimensions strengthen the interpretation that the estimates capture social ties rather than a single proxy such as nationality. The fact that arrival cohort matters argues against simple observable sorting by ethnicity or country of origin.

Economic message: Favoritism is a broad social-connection phenomenon in this workplace, not merely ethnic favoritism.

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TABLE A.I

SOCIAL CONNECTIONS, MANAGERIAL INCENTIVES, AND THE COO'S ALLOCATION ALGORITHM[a]

	Probability of Being Selected to Pick[b]	Probability of Being Unemployed[c]
Performance bonus for managers	1.34	2.04*
	(.495)	(.764)
Performance bonus for managers × worker i is socially connected	.524	.605
	(.214)	(.253)
Total yield in site 1	2.24***	.802***
	(.153)	(.057)
Total yield in site 2	.883***	.800***
	(.036)	(.032)
Number of workers available to pick fruit	.380***	1.83***
	(.037)	(.178)
Worker i's previous deviation from mean productivity	1.16*	1.07
	(.091)	(.107)
Log-likelihood	−5186.8	−3208.5
Number of observations (worker-day)	15,551	9808

[a]Standard errors are given in parentheses, clustered by worker. *** denotes that the log odds ratio is significantly different from 1 at 1%, ** at 5%, and * at 10% levels. Conditional logit estimates are reported where observations are grouped by worker. All continuous variables are divided by their standard deviations so that one unit increase can be interpreted as increase by one standard deviation. A manager and given worker i are defined to be connected if they are either of the same nationality, live in the same area, or are in the same arrival cohort. "Total yield" on the site is the total kilograms of the fruit picked on the site-day. The "number of workers available to pick fruit" is the total number of individuals that are on the farm that day and are available for fruit picking. "Worker i's previous deviation from mean productivity" is defined on the last day the worker was selected to pick. We first take the deviation of the worker's productivity from the field average productivity on each field he picked on the day he was last selected to

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TABLE A.II

SOCIAL CONNECTIONS AND MANAGERIAL INCENTIVES

	Type of Social Connect
Share of managers of same nationality as i, fixed wages for managers	.162***
	(.045)
Share of managers of same nationality as i, performance bonus for managers	−.075
	(.134)
Share of managers living in same area as i, fixed wages for managers	.087*
	(.049)
Share of managers living in same area as i, performance bonus for managers	−.070
	(.071)
Share of managers of same arrival cohort as i, fixed wages for managers	.309***
	(.088)
Share of managers of same arrival cohort as i, performance bonus for managers	−.079
	(.142)
Interactions of nationality × performance bonus dummy	Yes [.068]
Interactions of living site × performance bonus dummy	Yes [.000]
Interactions of arrival cohort × performance bonus dummy	Yes [.000]
Interactions of worker fixed effect × performance bonus dummy	No
Field-date fixed effects	No
Adjusted R-squared	.4366
Number of observations (worker-field-day)	8884

[a]Dependent variable = log of worker's productivity (kilograms picked per hour on the field-day). *** denc significance at 1%, ** at 5%, and * at 10%. Standard errors are reported in parentheses and allow for clustering the worker level. All specifications control for worker, field, and manager fixed effects. The other controls inclu the managerial performance bonus dummy, the worker's picking experience, the field life cycle, a time trend, a interactions between the performance bonus dummy and the worker's nationality, arrival cohort, and living site. T field life cycle is defined as the nth day the field is picked divided by the total number of days the field is picked o the season. All continuous variables are in logarithms. At the foot of the column we report the p-value on the F-t on the joint significance on the interaction terms with the performance bonus dummy.

8.9 Table A.III: Robustness of Results to Time Effects

Read:

- Columns add farm-specific, field-specific, and worker-specific time trends.
- The fixed-wage connection effect remains positive and significant.
- Placebo tests based on field life cycle and the 2004 season show similar pre/post patterns do not arise mechanically from time.
- The 2004 placebo is especially useful because managers were paid fixed wages throughout the entire season.

Detailed interpretation: The main concern is that the connection effect disappears after June 27 for reasons unrelated to incentives, such as seasonality, field depletion, or workforce learning. Table A.III shows that these patterns do not explain the estimates. In 2004, when no bonus was introduced, connections continue to matter across the analogous date.

Economic message: The evidence supports the interpretation that the bonus changed managerial behavior rather than the connection effect naturally fading over the season.

TABLE A.III
ROBUSTNESS OF RESULTS TO TIME EFFECTS[a]

	1. Farm Specific Time Trend	2. Field Specific Time Trend	3. Worker Specific Time Trend	4. Placebo Bonus Based on Field Life Cycle	5. Placebo Bonus Based on 2004 Season
Share of managers connected to i, fixed wages for managers	.165*** (.043)	.188*** (.067)	.269*** (.094)		
Share of managers connected to i, performance bonus for managers	−.037 (.092)	−.003 (.116)	.345 (.511)		
Share of managers connected to i, fixed wages for managers × 2nd quarter dummy (31st May)	−.018 (.076)				
Share of managers connected to i, performance bonus for managers × 4th quarter dummy (29th July)	−.133 (.099)				
Share of managers connected to i, fixed wages for managers × field life cycle		−.089 (.141)			
Share of managers connected to i, performance bonus for managers × field life cycle		−.249 (.200)			
Share of managers connected to i, fixed wages for managers × days on farm for worker i			−.047 (.035)		
Share of managers connected to i, performance bonus for managers × days on farm for worker i			−.109 (.132)		
Share of managers connected to i, placebo bonus based on field life cycle = 0				−.087 (.081)	
Share of managers connected to i, placebo bonus based on field life cycle = 1				−.033 (.138)	
Share of managers connected to i, placebo bonus 2004 = 0					.201* (.109)
Share of managers connected to i, placebo bonus 2004 = 1					.215*** (.033)
Interactions of nationality × performance bonus dummy	Yes	Yes	Yes	Yes	Yes
Interactions of living site × performance bonus dummy	Yes	Yes	Yes	Yes	Yes
Interactions of arrival cohort × performance bonus dummy	Yes	Yes	Yes	Yes	Yes
Adjusted R-squared	.4374	.4524	.4368	.6260	.4532
Number of observations (worker-field-day)	8884	8884	8884	1584	2692

[a]Dependent variable = log of worker's productivity (kilograms picked per hour on the field-day). *** denotes significance at 1%, ** at 5%, and * at 10%. Standard errors are reported in parentheses and allow for clustering at the worker level. All specifications control for worker, field, and manager fixed effects. The other controls included in each specification are the managerial performance bonus dummy, the worker's picking experience, the field life cycle, and a time trend. The field life cycle is defined as the nth day the field is picked divided by the total number of days the field is picked over the season. All continuous variables are in logarithms. A manager and given worker i are defined to be connected if they are either of the same nationality, live in the same area, or are in the same arrival cohort. All sample workers are connected to at least one manager on at least one field-day and work at least one week under each incentive scheme. In column 1 the 2nd quarter dummy is defined to be equal to 0 before May 31st and equal to 1 thereafter. The 4th quarter dummy is defined to be equal to 0 before July 29th and equal to 1 thereafter. These dummy variables split the pre- and postbonus periods equally into two halves. In column 3 the days on the farm for a worker are defined as the number of days elapsed since the worker first arrived on the farm. In column 4 the placebo bonus dummy based on the field life cycle is defined to be 0 if the field is less than .53 of the way though its life cycle, and 1 otherwise. In this column the sample is restricted to fields that are only operated in the period when managers are paid a performance bonus (after June 27th). In column 5, the sample covers the same period of time (May 1st to Aug 31st) in the following year—2004—when managers were paid wages throughout. The placebo bonus is equal to 1 after June 27, 2004. The interaction terms at the foot of the table are defined with respect to the placebo bonus dummy variable in columns 4 and 5.

9 Detailed result map

- **Social ties exist and vary within worker.** Table I shows average connectedness is similar across incentive regimes and has enough within-worker variation for identification.
- **Connected workers benefit under fixed wages.** Table II and Table III show connected workers are more productive when managers have low-powered incentives.
- **Performance pay removes the connection premium.** Table III shows social connection effects disappear when managers are paid bonuses.
- **The bonus changes allocation across ability.** Table IV and Figure 1 show that low ability connected workers lose productivity while high ability workers gain.
- **Firm-level productivity falls with social connections under fixed wages.** Table V shows average field productivity is lower when average connectedness is higher.
- **Assignment and time confounds are addressed.** Appendix Tables A.I-A.III show robustness to allocation rules, connection definitions, and time effects.

10 Short summary

- The paper studies social connections between managers and workers in a fruit-picking firm.
- Workers are paid piece rates and managers can target logistical help to individual workers.
- Social connections are measured using nationality, arrival cohort, and living proximity.
- The authors engineered a natural field experiment that introduced performance pay for managers.
- Under fixed wages, managers favor socially connected workers.
- Under performance bonuses, managers instead target high ability workers.
- The productivity premium for connected workers is about 9% when all managers on the field are connected to the worker.
- The premium disappears under managerial performance pay.
- Low ability connected workers lose productivity when bonuses are introduced, while high ability workers gain.
- Average field productivity is lower when social connectedness is higher under fixed wages.
- The evidence supports a negative allocation effect of social connections in this setting.
- The results do not imply that social ties are always harmful; in more complex tasks, communication and trust benefits may dominate favoritism costs.

11 Conclusion

11.1 Core conclusion

Bandiera, Barankay, and Rasul show that social connections inside firms can shape the allocation of managerial effort and the productivity of workers. In their setting, managers help socially connected workers when incentives are low-powered. Once managers receive bonuses tied to field productivity, they stop favoring connected low ability workers and instead target high ability workers. This reveals that the earlier allocation of managerial effort was not productivity maximizing.

11.2 Economic intuition

The key intuition is that managers face two motives. The first is a productivity motive: help the workers whose output responds most to managerial effort. The second is a social motive: help workers to whom the manager is connected. Fixed wages give the social motive more room to operate. Performance pay increases the return to the productivity motive, so managerial help is redirected toward high ability workers. The firm benefits from this reallocation even though some connected workers lose.

11.3 Why the evidence is persuasive

The design combines within-worker variation, manager fixed effects, a field experiment, field-date fixed effects, and placebo checks. This combination allows the authors to distinguish favoritism from worker ability, manager quality, field quality, and seasonality. The most persuasive fact is not simply that connected workers are more productive, but that the connection effect disappears exactly when managers are given a financial stake in overall productivity.

11.4 Organizational implications

The paper implies that firms should think jointly about social structure and incentives. Social ties may be valuable when workers need coordination, trust, or informal communication. But when tasks are simple and productivity is measurable, social ties may cause managers to allocate help inefficiently. Stronger performance pay can reduce favoritism, but may also weaken potentially useful relational mechanisms.

11.5 Limitations and external validity

The farm setting has unusually precise measurement, simple tasks, short-term employment, and observable managerial actions. Effects may differ in settings with complex teamwork, subjective evaluation, long-run relationships, or hidden forms of favoritism. The paper is therefore strongest as causal evidence on one organizational environment and as a framework for thinking about how social relations interact with incentives.

11.6 Takeaway

The incremental lesson is that social capital inside firms is not automatically productivity enhancing. Its value depends on the incentive system and the technology of managerial effort. In this workplace, social connections create favoritism under fixed wages, and performance pay disciplines managers to allocate effort more efficiently.